

FAIRVIEW AGDM, AB, Canada

WMO: 713470

Lat: 56.0814N

Lon: 118.4394W

Elev: 2147

StdP: 13.59

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.9	-20.4	-32.6	0.9	-25.9	-26.7	1.3	-20.2	27.4	37.2	23.9	36.6	3.6	30	0.698

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.9	81.4	60.9	77.9	59.3	74.9	58.1	63.7	76.1	61.7	74.3	60.0	71.7	8.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.0	80.6	68.7	56.9	74.8	65.5	55.2	70.2	63.6	30.1	76.6	28.6	74.5	27.4	71.8	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.3	19.5	17.0	DB	-33.2	86.7	5.6	3.1	-37.2	88.9	-40.4	90.7	-43.6	92.5	-47.6	94.7	(4)
(5)			WB	-33.7	67.1	5.8	3.1	-37.9	69.3	-41.3	71.1	-44.5	72.8	-48.7	75.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.5	10.3	11.3	20.8	36.9	50.2	57.5	60.8	58.0	49.5	38.1	19.9	10.8		
	DBStd	21.97	16.10	14.91	13.82	9.72	8.04	5.52	5.14	6.16	7.81	8.61	13.34	14.68		
	HDD50	6332	1231	1085	907	400	95	6	2	8	100	379	904	1215		
	HDD65	10818	1696	1505	1372	843	459	232	148	229	465	835	1354	1680		
	CDD50	1028	0	0	0	7	102	231	337	256	86	9	0	0		
	CDD65	40	0	0	0	0	1	7	19	12	1	0	0	0		
	CDH74	859	0	0	0	3	51	145	358	244	57	1	0	0		
	CDH80	155	0	0	0	0	7	21	67	54	6	0	0	0		
	WSAvg	7.5	6.7	6.5	7.8	8.8	9.0	8.0	7.3	7.1	7.8	7.9	6.7	6.5		
	PrecAvg	18.0	1.1	0.9	0.7	0.8	1.8	3.0	3.1	2.2	1.5	1.1	0.9	1.0		
(15) Precipitation	PrecMax	24.4	2.4	3.2	1.9	2.5	6.2	6.5	6.3	5.4	5.1	2.6	1.9	2.9		
	PrecMin	11.8	0.2	0.2	0.2	0.1	0.2	0.4	0.4	0.4	0.2	0.1	0.0	0.2		
	PrecStd	3.2	0.6	0.7	0.5	0.6	1.4	1.7	1.4	1.3	1.0	0.6	0.5	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.9	42.4	48.4	70.9	79.8	82.9	86.0	85.6	80.1	69.2	51.4	42.1		
		MCWB	37.6	36.0	40.4	50.4	55.7	60.9	63.5	61.8	58.7	52.8	42.0	36.6		
	2%	DB	39.9	39.0	44.2	61.2	73.6	77.9	81.4	80.5	74.5	59.9	45.1	39.0		
		MCWB	35.4	34.2	37.1	45.3	52.7	58.2	61.9	61.0	56.9	47.6	38.9	34.4		
	5%	DB	37.3	35.7	41.2	55.7	69.6	74.6	77.9	75.9	68.3	54.7	40.8	35.0		
		MCWB	33.7	32.0	35.2	42.6	51.1	57.1	60.6	59.0	54.3	44.6	36.1	31.6		
	10%	DB	33.6	32.0	38.5	51.4	65.8	71.1	74.7	72.3	63.6	51.1	37.0	31.4		
		MCWB	30.8	28.5	33.4	40.5	49.2	55.7	59.5	57.9	51.6	42.7	33.1	28.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.1	36.9	41.0	51.0	57.9	63.6	67.5	65.0	60.8	53.9	43.0	37.6		
		MCDB	41.8	41.4	47.2	69.2	74.7	77.4	78.8	77.5	75.4	67.5	49.8	41.3		
	2%	WB	36.0	34.6	37.8	46.6	54.4	60.8	64.6	62.8	57.9	48.8	39.4	34.7		
		MCDB	39.4	38.4	43.6	60.3	69.6	73.7	75.8	76.3	70.8	58.0	44.1	38.3		
	5%	WB	34.1	32.3	35.7	43.3	52.2	58.8	62.6	60.9	55.3	45.7	36.5	32.0		
		MCDB	37.3	36.2	40.8	54.1	66.4	71.0	74.7	73.4	67.0	52.7	40.5	35.0		
	10%	WB	30.7	29.0	33.7	41.1	50.5	56.9	60.8	58.9	52.8	43.3	33.4	28.7		
		MCDB	34.0	32.4	38.2	50.5	63.4	68.6	72.2	69.8	62.0	49.9	36.5	31.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.0	17.5	18.2	18.4	22.3	21.4	22.9	23.1	21.4	17.3	15.5	15.4		
		MCDBR	16.1	18.7	18.6	25.8	28.2	27.8	29.3	29.6	28.7	23.5	17.3	17.7		
		MCWBR	12.7	14.7	12.8	13.8	13.1	12.6	13.0	13.0	13.8	13.5	12.6	14.9		
		MCDBR	15.3	18.2	17.5	24.1	25.8	23.9	26.0	26.4	27.1	22.2	16.3	17.5		
	5% WB	MCWBR	12.6	14.5	12.4	13.5	12.6	11.9	13.3	13.0	13.6	13.7	12.4	15.0		
(40) Clear-Sky Solar Irradiance	taub		0.259	0.249	0.296	0.320	0.341	0.350	0.351	0.348	0.308	0.283	0.241	0.223		
	taud		2.177	2.341	2.278	2.387	2.399	2.393	2.406	2.406	2.526	2.548	2.405	2.223		
	Ebn at Noon		205	269	277	286	285	283	280	274	273	255	225	197		
	Edh at Noon		18	22	31	32	34	35	34	32	24	18	14	13		
(44) All-Sky Solar Radiation	RadAvg		222	526	1004	1491	1772	1911	1857	1511	1035	548	244	154		
	RadStd		23	41	76	81	122	155	112	133	87	64	21	12		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page